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PROJECT REPORT: WATERWISE SMART IRRIGATION

Prepared by fictional team Cedar Lab for a classroom demonstration. The project explores low-cost soil monitoring for a community garden.

Problem: Garden volunteers water on a fixed schedule even after rain. This can waste water and leave some planting beds too wet.

Approach: A capacitive soil-moisture sensor sends a reading to a small gateway every fifteen minutes. The dashboard stores readings and suggests irrigation only when moisture falls below the selected threshold.

Prototype: Four fictional plots were monitored for six weeks. The demonstration dataset showed a median 18 percent reduction in scheduled watering compared with the baseline calendar.

Evaluation: Sensor placement, calibration, and shaded beds affected the readings. The result is a classroom prototype, not a field-validated agricultural product.

Next steps: Add a rain gauge, log calibration changes, and let a gardener confirm each irrigation recommendation.

Sample data notice: Metrics, names, and observations are fictional and intended only for demonstrating search and document processing.